

Sustainable Distributed Adaptive Platoon in Multi-Agent Mobile-Edge Computing Networks for Lane Reduction Scenario

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Abstract—Nowadays, Connected Automated Vehicles (CAVs) have emerged as powerful infrastructures for the next-generation Intelligent Transportation System (ITS) as the rapid technological advancements of communication networks and vehicular intelligence. While prospective platoon-based techniques in CAVs, the heterogeneous traffic condition poses a challenge for platoon control in the self-organized traffic bottleneck, thus making an urgent need for a practical sustainable transportation architecture. To address this problem, we propose a software defined architecture that leverages multi-agent techniques to mobile-edge computing networks for multi-vehicle adaptive platoon, which is called SD-M3ASP. **The architecture supports centralized and decentralized management of vehicular edge communication resources between vehicles and edge devices.** Then, we propose cluster-based kinematic models by grouping vehicles into multi-vehicle clusters (MVCs) to facilitate efficient platoon control with collision avoidance. Furthermore, we propose three-stage platoon control algorithms to adaptively balance the size of MVCs and form stable platoons. The intra-platoon and inter-platoon convergence are analyzed by using the Routh stability criterion and Lyapunov technique. A CAV simulation software is developed for demonstration purposes which is available online at <http://qgailab.com/cav-sim>. Extensive numerical simulation results have shown the superiority of the proposed method, which can greatly eliminate the self-organized congestion caused by heterogeneous traffic flow.

Index Terms—connected and automated vehicles, multi-agent systems, mobile-edge computing, adaptive platoon control, leader-follower consensus, lane reduction.

I. INTRODUCTION

THE most common phenomenon in traffic flow is traffic congestion, which has a direct impact on the waste of energy consumption and additional environmental pollution [1], [2]. By 2050, it is estimated that 70% of the world's population will live in cities, which will increase the pressure on urban infrastructures and transportation systems [3]. Among the

resulting problems, traffic congestion will continue to be one of the most urgent problems. Zheng [4] studied the relationship between crash occurrence likelihood and traffic conditions and found that the probability of accidents in congested traffic flow is six times of that in free traffic flow. According to the recent report by Mekker et al. [5], the congested crash rate is even worse (25 times).

Connected and Automated Vehicles (CAVs) are the key infrastructures of the next-generation Intelligent Transportation System (ITS), which have great potential to predictably reduce congestion delays through vehicular communication of exchanging information about traffic conditions. They also have great promises to reduce redundant energy consumption and emissions [6], [7]. Besides, the platoon-based driving pattern in CAVs has a significant potential to avoid the frequent stops and delays, and ensure safely driving, which is referred to as vehicles formed into a unified string with harmonized velocities and safe inter-vehicle distances [8], [9]. Zhou et al. [10] studied the effect of platoon size of CAVs on capacity and flow stability. Wu et al. [11] designed an adaptive distributed event-triggered controller to achieve the stability of vehicular platoon with Radial basis function neural networks-based approximation of the uncertain function. Thormann et al. [12] proposed a distributed model predictive control for safe and efficient platooning vehicles with dense-spacing allowance and loose communication requirements.

The inhomogeneity of traffic road, especially in lane closings, intersections or uphill gradients, is easy to cause congested traffic flow [13], [14]. Of particular concern is the prototypical traffic bottleneck as shown in Fig. 1, which is a 2-to-1 lane reduction scenario. Vehicles in the terminated lane must change to the continuous lane before reaching the end of the lane (e.g., p_e in Fig. 1). Congestion in lane reduction can be self-organized because of drivers' aggressive lane change and lane squeeze [15], which is also called emergency evacuation planning [16]. The evidence in [17] suggested that early lane changes from the terminated lane to the continuous lane can eliminate congestion and reduce travel times past the bottleneck, depending on the amount of incoming traffic. When traffic density is relatively high, drivers are forced to decelerate and accelerate abruptly and frequently. On highways, even without signal interruptions, stop-and-go traffic patterns can be formed and propagated backward along the shockwave, which undermines the performance of transportation systems [18]. Therefore, the heterogeneity of

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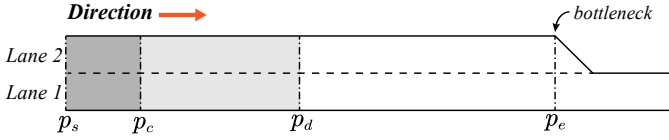


Fig. 1: Illustration of the 2-to-1 lane reduction scenario

headways and gaps between successive vehicles is essentially posing a challenge for platooning techniques that are more suited for homogeneous traffic conditions.

This study mainly focuses on the heterogeneous traffic congestion problem in the prototypical 2-to-1 lane reduction scenario. The main contributions of this paper are threefold:

- **Architecture:** In order to support the deployment and management of the practical vehicular networks, we propose a Software Defined architecture that leverages Multi-agent technique to Mobile-edge computing networks for Multi-vehicle Adaptive Switching Platoon, which is called **SD-M3ASP**. The proposed architecture provides supports for both centralized and decentralized management of vehicular edge communication resources between vehicles and edge devices. Additionally, we provide the corresponding triple-stage platoon control workflow of SD-M3ASP in the 2-to-1 lane reduction scenario, which includes the topology construction stage, adaptive topology switching stage, and adaptive platoon control stage.
- **Model:** As the heterogeneous inter-vehicle gap and velocity, single leader oriented platoon models are inadequate for multi-lane and long-range platoon control, **which inevitably leads to a low space utilization ratio**. Besides, attributable to the limitation of vehicle communication distance, tail follower vehicles are challenging to communicate with the head leader in the long-range platoon. Therefore, it is more suitable to split vehicles into multiple clusters, thus formulating cluster-based platoons. **Therefore**, we propose two kinds of leader-follower kinematic models based on Multi-vehicle Clusters (MVCs). Specifically, the head and rear leader vehicles are considered to guide follower vehicles within the cluster to form a stable platoon. Besides, leader vehicles can coordinate with each other according to the collision avoidance vehicles. The theoretical analyses of inter- and intra-platoon convergence are carried out by using the Routh stability criterion and the Lyapunov technique.
- **Algorithm:** **The time-varying and heterogeneous traffic flow density is a common condition that has a considerable impact on the design of platoon control algorithms**. On the basis of three stages in the aforementioned architecture SD-M3ASP, the algorithms operating under SD-M3ASP are proposed, which initialize MVCs, switch and adaptively select leader and form a platoon formation based on MVCs. Specifically, we initialize MVCs and determine the leader of each MVC in the Topology Construction Stage. **Then, we stipulate that vehicles between MVCs can be redistributed in the Topology Switching**

TABLE I: Abbreviation

Abbreviation	Explanation
ITS	Intelligent Transportation System
CAV	Connected Autonomous Vehicle
SDN	Software Defined Network
DAI	Distributed Artificial Intelligence
MAS	Multi-agent System
MEC	Mobile-edge Computing
RSU	Road-side Unit
OBU	On-board Unit
V2I	Vehicle to Infrastructure
V2V	Vehicle to Vehicle
V2X	Vehicle to Everything
MVC	Multi-vehicle Cluster

phase, which preprocesses and balances MVCs according to real-time road capacity. Finally, we adaptively determine the state of the leader vehicle, thus forming stable platoons by using aforementioned kinematic models in the Platoon Control Stage. We demonstrate that the self-organized congestion at the lane reduction bottleneck can be effectively eliminated by taking advantage of the proposed algorithms.

The remainder of this paper is structured as follows. Section II briefly summarizes and classifies the related work about CAVs. Section III provides necessary background materials. Section IV designs SD-M3ASP architecture and the corresponding workflow. Section V elaborates kinematic models and platoon control algorithms in SD-M3ASP. Section VI analyzes the intra- and inter-platoon convergence. Section VII provides the numerical simulations to back up our theory. Section VIII concludes this paper and discusses future work.

II. LITERATURE REVIEW

The development of CAVs is largely influenced by the progressive prosperity of Software Defined Network (SDN) [19], 5G [20], cloud-edge infrastructure [21] and Distributed Artificial Intelligence (DAI) [22], we are approaching the apogee of smart Intelligent Transportation System (ITS). The latest developments in these fields have shown an increased interest in the deployment of CAVs:

a) **CAVs with Software Defined Network:** The basic principle of SDN is to separate the control plane from the data plane to achieve the purpose of decoupling [23], [24]. Generally, vehicle communication is based on two main technologies: dedicated short-range communication and cellular network [25]. By using these two technologies, vehicular information, obtained from vehicular sensors, other adjacent vehicles, road side units (RSU), can be rapidly passing on the vehicular network. Accordingly, vehicular network can provide a variety of complex delay-sensitive applications for a large number of vehicles [26], such as autonomous driving [27], task offloading [28], path planning [29], etc. Meanwhile, SDN is fast becoming a key instrument to satisfy such various ever-increasing vehicular application demands, which can manage the vehicular communications centrally and provide suitable and efficient communication flows.

TABLE II: Nomenclature

Symbol	Definition
D_c	the maximum communication distance of V2I
F	the longitudinal secure distance between two vehicles
Z	the longitudinal length of the vehicle
$\mathcal{H}, h$	head leader and head follower of the MVC
$\mathcal{R}, r$	rear leader and rear follower of the MVC
$\mathbf{A} = [a_{ij}] \in \mathbb{R}^{n \times n}$	the adjacency matrix between follower vehicles
$\mathbf{B} = [b_{i\mathcal{H}}] \in \mathbb{R}^n$	the adjacency matrix between the head leader $\mathcal{H}$ and follower vehicles i
$\mathbf{C} = [c_{i\mathcal{R}}] \in \mathbb{R}^n$	the adjacency matrix between the rear leader $\mathcal{R}$ and follower vehicles i
$\mathbf{P} = [p_{ij}] \in \mathbb{R}^{n \times n}$	the desired longitudinal gap matrix between follower vehicles i, j
$\mathbf{Q}_{\mathcal{H}} = [q_i] \in \mathbb{R}^n$	the desired longitudinal gap matrix between the head leader $\mathcal{H}$ and follower vehicles i
$\mathbf{Q}_{\mathcal{R}} = [r_i] \in \mathbb{R}^n$	the desired longitudinal gap matrix between the rear leader $\mathcal{R}$ and follower vehicles i
CS	the construction set of the MVC
$ CS $	the size of the construction set
CS_i	the i -th vehicle in the CS
p_i^{CS}	$i = 0$ indicates the head leader vehicle, otherwise follower vehicles; $i = -1$ indicates the last vehicle in the CS .
DS	the longitudinal position of the i -th vehicle in the CS
$ DS $	the decision set of MVCs; $DS = \{M^1, \dots, M^n\}$
M^k	the size of the decision set
$ M^k $	the sub-decision set in the DS ; $M^k = \{M_0^k, M_1^k, \dots, M_n^k\}$
$\mathbf{D} = [d_{LL'}] \in \mathbb{R}^{ DS -1}$	the size of the sub-decision set
$\mathbf{E} = [e_{Lm}] \in \mathbb{R}^{ DS }$	the adjacency matrix between head leaders L and their front head leader vehicles L'
$\bar{n}$	the adjacency matrix between head leaders L and their corresponding avoidance vehicles m
$\bar{n}$	the maximum allowable capacity of the MVC
$\bar{n}$	the average cluster capacity of MVCs

b) CAVs with Mobile Edge Computing: Cloud computing has been applied in various fields, where user data needs to be transmitted to and processed in the cloud center. However, the cloud center is usually far away from vehicles, which may not provide guarantees for low-latency vehicular applications, thus becoming a bottleneck in deploying vehicular networks [28]. Besides, it is not feasible or economical to transfer large amounts of data from vehicles to the remote cloud center. In order to solve these problems, Mobile Edge Computing (MEC) is considered as a promising solution to deploy additional computing capacities closer to vehicles, which can effectively reduce latency and improve the utilization of cache resources [30], [31]. By deploying MEC servers in RSUs near the road, resource-constrained vehicles can meet the requirement of low latency [28], [32]. However, most studies [17], [33], [34] on the lane reduction scenario are based on ideal assumptions of vehicular communication network and computing capacity, which should be properly dimensioned.

c) CAVs with Distributed Artificial Intelligence: The Multi-agent System (MAS) is one of the most significant innovations in Distributed Artificial Intelligence (DAI), in which a group of autonomous and cooperative agents seeks to achieve consensus in a distributed fashion [35]. The platoon control in CAVs can be solved from the perspective of MAS, where each CAV can be regarded as an agent. Bernardo et al. [36] provide a distributed control protocol to form a platoon in the presence of heterogeneous time-varying delays. Oliveira et al. [37] design a consensus-based platoon control solution with variable communication topologies. Li et al. [9] study the consensus-based platoon control under the Vehicle-to-Everything (V2X) communications. However, most studies mainly focus on the single-platoon performance of CAVs. There has been little discussion about multi-lane and multi-platoon problems [38] and the selection of leader vehicles

[39] which is inevitable in the lane reduction scenario, thus remaining unclear and need further systematically studied.

Overall, it is not satisfactory that a single technology can support such a variety of vehicular applications. There is still a lack of work studying and leveraging emerging technologies into the practical design of transportation systems in the 2-to-1 lane reduction scenario, especially with dynamic heterogeneous traffic condition.

III. PRELIMINARIES

For convenience, the main abbreviations and nomenclatures used in this paper are summarized in TABLE I and II.

Suppose that a vehicular platoon is composed of one leader vehicle (labeled as vehicle 0) and n follower vehicles (labeled as vehicles $1, 2, \dots, n$). In this paper, we use a directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ to describe the communication topology between vehicles, where $\mathcal{V} = \{0, 1, \dots, n\}$ is a set of vehicles, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is a set of edges. A directed edge $(j, i) \in \mathcal{E}$ means that there is a connection from vehicle j to vehicle i , indicating that vehicle i can obtain the state information from vehicle j . Then, we characterize the weighted vehicular communication network of follower vehicles in graph $\mathcal{G}$ by the adjacency matrix $\mathbf{A} = [a_{ij}] \in \mathbb{R}^{n \times n}$. Here, a_{ij} indicates that vehicle i can obtain the state information of vehicle j via the vehicular network, i.e., $a_{ij} > 0$ if $(j, i) \in \mathcal{E}$; otherwise $a_{ij} = 0$. Notably, we assume that there is no self-loop in $\mathcal{G}$, i.e., $a_{ii} \neq 0$. Then, the neighbor set of vehicle i can be described as $\mathcal{N}_i = \{j \mid a_{ij} > 1\}$.

The Laplacian matrix $\mathbf{L} = [l_{ij}] \in \mathbb{R}^{n \times n}$ of follower vehicles in graph G is defined as $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ is the indegree matrix with elements $d_i = \sum_{j \neq i}^n a_{ij}$ and zero off-diagonal elements. Therefore, $l_{ij} = -a_{ij}$ if $i \neq j$; otherwise, $l_{ij} = \sum_{j=1}^n a_{ij}$. Then, we

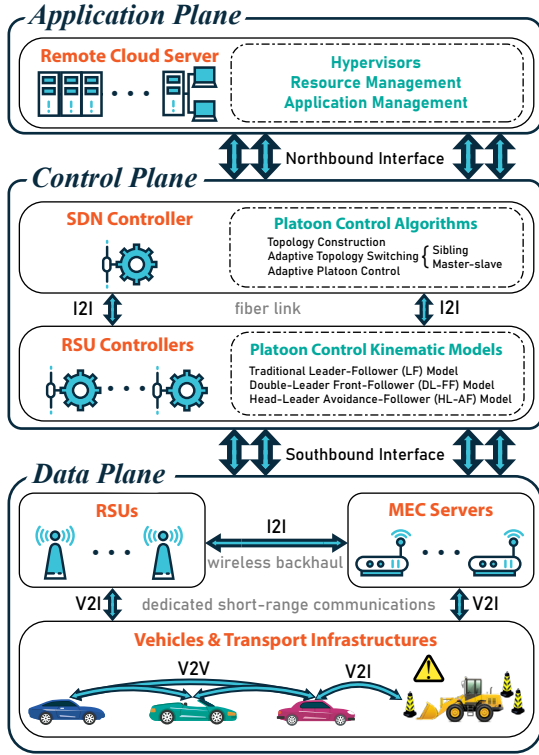


Fig. 2: The conceptual architecture SD-M3ASP

denote by $\mathbf{P} = [r_{ij}] \in \mathbb{R}^{n \times n}$ the desired longitudinal gap matrix between follower vehicles, where $r_{ij} = (j-i) * (F+Z)$, and F is the longitudinal secure distance. Then, we denote by $\mathbf{Q} = [r_i] \in \mathbb{R}^n$ the desired longitudinal gap matrix between the leader vehicle and n follower vehicles, where $r_i = -i * (F+Z)$.

With reference to the vehicular kinematics model [40], this paper designs the controller based on the leader-follower approach. The controller for follower vehicle i at time t is defined as

$$\begin{cases} \dot{\mathbf{x}}_i(t) = \mathbf{v}_i(t) \\ \dot{\mathbf{v}}_i(t) = \mathbf{u}_i(t) \end{cases} \quad (1)$$

where $\mathbf{x}_i(t) = [p_i(t), q_i(t)]^T$, $\mathbf{v}_i(t) = [\omega_i(t), \theta_i(t)]^T$, $\mathbf{u}_i(t) = [\chi_i(t), \mu_i(t)]^T$ are the position, velocity and control input of follower vehicle i , $i = 1, \dots, n$.

The controller for leader vehicle L at time t is defined as

$$\begin{cases} \dot{\mathbf{x}}_L(t) = \mathbf{v}_L(t) \\ \dot{\mathbf{v}}_L(t) = \mathbf{u}_L(t) \end{cases} \quad (2)$$

where $\mathbf{x}_L(t) = [p_L(t), q_L(t)]^T$, $\mathbf{v}_L(t) = [\omega_L(t), \theta_L(t)]^T$, $\mathbf{u}_L(t) = [\chi_L(t), \mu_L(t)]^T$ are the position, velocity and control input of leader vehicle L , $L = 0$.

IV. ARCHITECTURE AND WORKFLOW

In this section, we present an overview of the proposed SDN-based platoon control architecture SD-M3ASP and elaborate the workflow of SD-M3ASP.

Workflow 1: Workflow of SD-M3ASP

- 1 Generate the MVC-based topology via Algorithm 1 ;
- 2 Obtain the adaptive switched MVCs according to the real-time road condition via Algorithm 2 (Algorithm 3 or Algorithm 4) ;
- 3 Establish MVC-based stable platoons in the continuous lane via Algorithm 5 ;

A. SDN-based Platoon Control Architecture in Multi-agent Mobile-edge Computing Networks

For clarity, Fig. 2 depicts the proposed conceptual platoon control architecture SD-M3ASP. SDN and MEC are leveraged in SD-M3ASP to effectively manage network resources and provide low latency services to CAVs. SD-M3ASP consists of three parts, i.e., application plane, control plane and data plane. The components and responsibilities of each plane in Fig. 2 are described in detail as follows.

- **Application Plane:** is the top layer of SD-M3ASP which is composed of a set of hypervisors to provide resource and application management. The resource and application management are primarily responsible for service generation and migration. The services of the application layer mainly include coordinating multiple vehicles to improve vehicular performance or user experience. In this paper, we focus on the platoon control services in the lane reduction scenerio. Other services may include security service, entertainment service, etc. According to the service requirements of users and vehicles, the rules and policies (i.e., platoon control algorithms & kinematic models) in the multi-agent and edge-computing network are generated and forwarded to the control plane.
- **Control Plane:** is the middle layer of SD-M3ASP which is composed of a central SDN controller and multiple RSU controllers. The central SDN controller provides global network management and decision making management according to the information collected by RSU controllers, which includes a set of platoon control algorithms. RSU controllers are mounted on RSUs near the road, which are responsible for forwarding the rules from SDN controller, which includes a set of platoon control kinematic models. SDN controller communicates with RSU controllers through fiber link.
- **Data Plane:** is the bottom layer of SD-M3ASP which is composed of RSUs, MEC servers, vehicles and other transport infrastructures. The vehicles are equipped with V2X devices (i.e., On-board Unit). RSUs can transmit vehicular states (e.g., location, velocity, etc.), road condition information from transport infrastructures (i.e., traffic signaling, traffic sign, etc.), and decision instructions of platoon control (e.g., lane changing, merging, diverging, etc.) between MEC servers and vehicles. The centralized SDN controller and distributed RSU controllers are responsible for managing this plane.

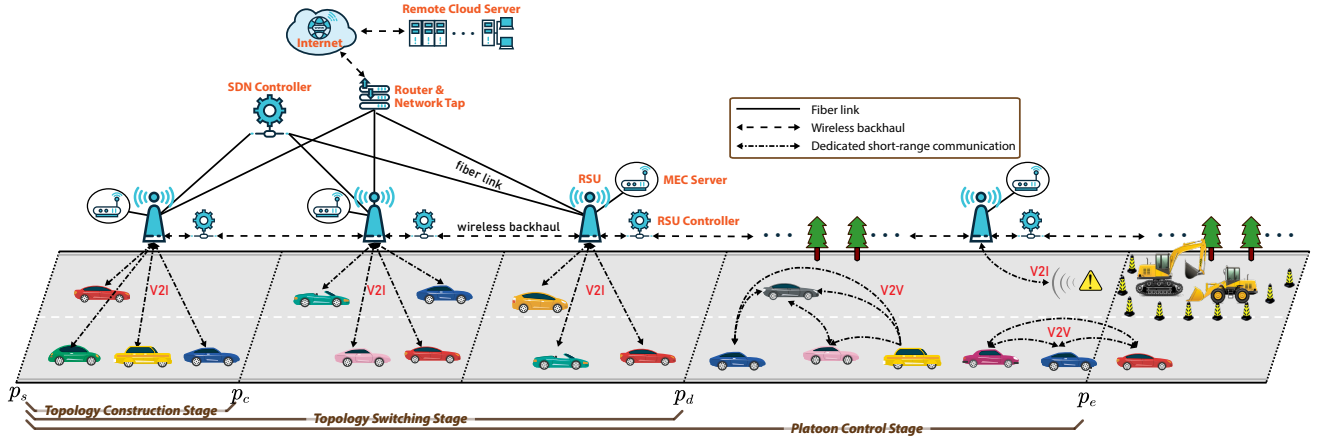


Fig. 3: The physical deployment of the SD-M3ASP architecture

B. Workflow of the SD-M3ASP architecture

For clarity, Fig. 3 depicts a physical deployment solution of SD-M3ASP, which is an actual scene at a specific time in the 2-to-1 lane reduction scenario. Besides, the workflow of SD-M3ASP is illustrated in Fig. 4 and Workflow 1.

The main idea of the proposed workflow is to form a stable platoon on the continuous lane before the end of the terminated lane (i.e., p_e), which can greatly eliminate congestion and reduce travel times past the bottleneck. p_s is the starting position of the SD-M3ASP workflow, and p_e is the end of the terminated lane. The ranges of $[p_s, p_c]$, $[p_s, p_d]$ and $[p_s, p_e]$ are the stage of ‘Topology Construction’, ‘Topology Switching’ and ‘Platoon Control’, respectively. In the stages of Topology Construction and Topology Switching, vehicles can communicate with RSUs via V2I to obtain road condition information, **can prepare** for the minimum unit of platoon control (MVC) through MEC servers in advance. Thereby, in the stage of Platoon Control, vehicles asymptotically and autonomously form a platoon according to adjacent MVCs through V2V communication.

Then, the detailed workflow in the range $[p_s, p_e]$ is described as follows.

1) *Topology Construction Stage*: is responsible for initializing the MVC in preparation for subsequent stages, where the MVC is the basic decision-making unit of platoon control. Specifically, when vehicle v_i passes the location p_s , v_i will send the topology construction request to the nearest RSU through its OBU (Step V1 in Fig. 4). The RSU forwards the request datagram to MEC server according to the rules of RSU controller. Then, MEC server determines v_i whether it is a leader vehicle according to the history cache of traffic flow (Step M1 in Fig. 4). After that, all vehicles entering after the leader will join the existing MVC as follower vehicles unless the leader vehicle passes through p_c (Step M2 in Fig. 4). RSUs will retain the information of each MVC, such as the physical address, location, velocity, acceleration, etc., for the subsequent maintenance and management of the vehicular cluster. We summarize the above steps in the stage of Topology Construction in Algorithm 1, and then elaborate them in Section V-B1.

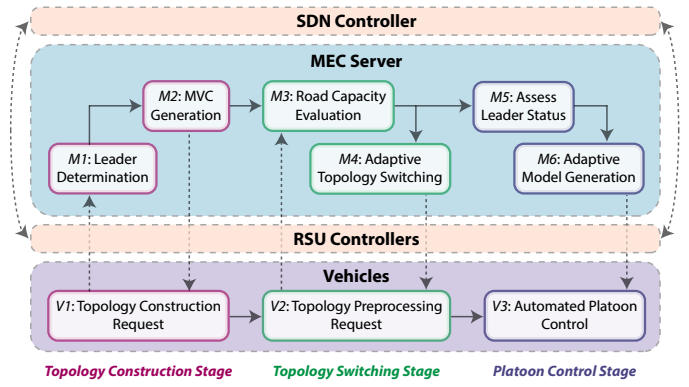


Fig. 4: The workflow of the SD-M3ASP architecture

2) *Topology Switching Stage*: is responsible for the adaptive preprocessing of MVCs according to traffic flow capacity. After the stage of Topology Construction, several MVCs have been constructed for vehicles within the range of p_s to p_c through the MEC server. When the leader vehicle in an MVC where vehicle v_i is located passes p_d , then the leader vehicle will transmit the overall information of all MVCs to the MEC server closest to p_d to make a topology preprocessing request (Step V2 in Fig. 4). The MEC server will first evaluate the road condition based on the information of all existing MVCs (Step M3 in Fig. 4). Then, the MEC server adaptively preprocess MVCs by switching vehicles in them according to the current road condition (Step M4 in Fig. 4). We summarize the above steps in the stage of Topology Switching in Algorithm 2, and then elaborate them in Section V-B2.

Remark 1: The distance between p_s and p_c directly affects the upper limit of vehicles in an MVC. The shorter the distance between p_s and p_c , the lower the upper limit of vehicles in an MVC. Besides, we provide an MVC-based metric to fully evaluate the real-time road condition. For this reason, the distance between p_s and p_d is empirically set to n times that between p_s and p_c , where $n \in \mathbb{N}^+$. Besides, the number of vehicles in the range $[p_s, p_d]$ is positively related to the number of deployed RSUs and the communication range of RSUs and vehicles. Accordingly, the selection of p_s is typically related to

physical properties of RSUs and vehicles in the actual scene.

3) *Platoon Control Stage*: is responsible for the implementation of adaptive platoon control. Vehicles in the terminated lane have to form a stable platoon to the continuous lane within the range of $[p_s, p_e]$. Specifically, a leader vehicle in an MVC needs to assess its status (i.e., accelerates, decelerates, or keeps a constant speed) according to the condition of the front adjacent MVC and of itself (Step *M5* in Fig. 4). Then, consensus-based intra-platoon and inter-platoon kinematic models are generated according to processed MVCs (Step *M6* in Fig. 4). Finally, vehicles are able to form stable platoons autonomously via V2V communication (Step *V3* in Fig. 4). We summarize the above steps in the stage of Platoon Control in Algorithm 3 and 4, and then elaborate them in Section V-B3.

Remark 2: In the design of SD-M3ASP, distributed MEC servers need to transmit the information of MVCs to the neighbor MEC server in the forward direction according to the driving direction, and keep them synchronized among MEC servers, thus guaranteeing adaptive decisions according to the real-time road condition.

In summary, the proposed SD-M3ASP architecture combines multi-agent technique and edge-computing network through SDN. From p_s to p_d , vehicles communicate with MEC servers through the edge-computing network of V2I to generate MVCs which are minimum decision-making units for platoon control; From p_s to p_e , vehicles form stable platoons through the multi-agent network of V2V according to the consensus-based kinematic models. It is a sustainable distributed architecture, which mainly solves the problem of adaptive platoon control for the scenario of lane reduction.

V. KINEMATIC MODELS AND ALGORITHMS

In this section, we present platoon control kinematic models and algorithms under the multi-agent mobile-edge computing communications context in Section V-A and V-B, respectively, which are corresponding to the three stages in the above proposed architecture SD-M3ASP.

A. Platoon Control Kinematic Models in SD-M3ASP

This section elaborates on the intra-platoon and inter-platoon models.

1) Design of consensus-based intra-platoon model:

a) *Traditional Leader-Follower Model*: Following the model used in [40], the basic leader-follower model for vehicle i in the k -th MVC is designed as

$$\begin{cases} \dot{x}_i^k(t) = v_i^k(t) \\ \dot{v}_i^k(t) = - \sum_{j=1}^N a_{ij}^k [\alpha (x_i^k(t) - x_j^k(t) - r_{ij}^k) + \beta (v_i^k(t) - v_j^k(t))] \\ \quad - [\gamma (x_i^k(t) - x_{\mathcal{H}}^k(t) - r_{i\mathcal{H}}^k) + \sigma (v_i^k(t) - v_{\mathcal{H}}^k(t))] \end{cases} \quad (3)$$

where a_{ij}^k is the adjacency weight between follower vehicle i and follower vehicle j in the k -th MVC; $r_{ij}^k = [r_{ij_p}^k, r_{ij_q}^k]^T$ is the desired gap between follower vehicle i and follower vehicle j in the k -th MVC, $r_{ij_p}^k, r_{ij_q}^k$ are the desired longitudinal gap and lateral gap; $r_{i\mathcal{H}}^k = [r_{i\mathcal{H}_p}^k, r_{i\mathcal{H}_q}^k]^T$ is the desired gap between follower vehicle i and leader vehicle $\mathcal{H}$, which has

a similar definition as r_{ij}^k ; $\alpha = [\alpha_p, \alpha_\omega]^T$, $\beta = [\beta_p, \beta_\omega]^T$, $\gamma = [\gamma_p, \gamma_\omega]^T$ and $\sigma = [\sigma_p, \sigma_\omega]^T$ are the gain parameters.

b) *Double Leader-Front Follower Model*: Following the model used in [38], [41], [42], we consider the situation that the state of the head leader and the rear leader in the MVC is relatively stable (i.e., the distance between them fluctuates little). Then, the follower vehicles in the middle are limited between the head and rear leader vehicles, thus having a better guarantee for safe collision-free platoon control. Therefore, we propose a new Double Leader-Front Follower (DL-FF) model, wherein the longitudinal model for follower vehicle i in the k -th MVC is designed as

$$\begin{cases} \dot{p}_i^k(t) = \omega_i^k(t) \\ \dot{\omega}_i^k(t) = - \sum_{j=1}^N a_{ij}^k [\phi_1 (p_i^k(t) - p_j^k(t) - r_{ij_p}^k) \\ \quad + \zeta_1 (\omega_i^k(t) - \omega_j^k(t))] \\ \quad - b_{i\mathcal{H}}^k [\phi_2 (p_i^k(t) - p_{\mathcal{H}}^k(t) - r_{i\mathcal{H}_p}^k) \\ \quad + \zeta_2 (\omega_i^k(t) - \omega_{\mathcal{H}}^k(t))] \\ \quad - c_{i\mathcal{R}}^k [\phi_3 (p_i^k(t) - p_{\mathcal{R}}^k(t) - r_{i\mathcal{R}_p}^k) \\ \quad + \zeta_3 (\omega_i^k(t) - \omega_{\mathcal{R}}^k(t))] \end{cases} \quad (4)$$

, and the lateral model for follower vehicle i in the k -th MVC is designed as

$$\begin{cases} \dot{q}_i^k(t) = \theta_i^k(t) \\ \dot{\theta}_i^k(t) = - b_{i\mathcal{H}}^k [\phi_4 (q_i^k(t) - q_{\mathcal{H}}^k(t) - r_{i\mathcal{H}_q}^k) + \zeta_4 (v_i^k(t) - v_{\mathcal{H}}^k(t))] \end{cases} \quad (5)$$

where $b_{i\mathcal{H}}^k$ and $c_{i\mathcal{R}}^k$ are the adjacency weight between follower vehicle i and head leader $\mathcal{H}$, rear leader $\mathcal{R}$ in the k -th MVC, respectively; $\phi = [\phi_1, \phi_2, \phi_3, \phi_4]^T$ and $\zeta = [\zeta_1, \zeta_2, \zeta_3, \zeta_4]^T$ are the gain parameters. The meanings of other notations are similar to those defined in formula (3).

Remark 3: In our architecture, the first vehicle of each MVC is the leader vehicle, and the rear leader vehicle mentioned here is actually the leader vehicle of the latter MVC. Specifically, the follower vehicle of the k -th MVC will not only refer to the state of the k -th MVC leader vehicle but also to the state of the $k+1$ -th MVC leader vehicle. This is the main difference between our kinematic model and the traditional model, it makes the vehicle platoon more stable and reliable.

2) *Design of consensus-based inter-platoon model*: In order to form a stable platoon and prevent collision between platoons of MVCs, we follow the model used in [38] and propose a new Head Leader-Avoidance Follower (HL-AF) model, wherein the longitudinal model for leader vehicle L in the k -th MVC is designed as

$$\begin{cases} \dot{p}_L^k(t) = \omega_L^k(t) \\ \dot{\omega}_L^k(t) = - d_{LL'}^k [\psi_1 (p_L^k(t) - p_L^{k-1} - r_{LL'_p}^k) + \xi_1 (\omega_L^k(t) - \omega_L^{k-1}(t))] \\ \quad - e_{Lm}^k [\psi_2 (p_L^k(t) - p_m^o - r_{Lm_p}^k) + \xi_2 (\omega_L^k(t) - \omega_m^o(t))] \end{cases} \quad (6)$$

Algorithm 1: Topology Construction

Input: vehicle v , CS
Output: CS , DS

```

1 if  $p_{\mathcal{H}}^{CS} > p_c$  then
2    $k = |DS| + 1$ ;
3   Append the set  $CS$  to the set  $DS$  as  $M^k$ ;
4   Delete the vehicles in the set  $CS$ ;
5    $|CS| = 0$ ;
6 end
7 if  $|CS| \neq 0$  then
8   Send keep-driving commands to vehicles in  $CS$ ;
9 end
10 if  $v$  is not Null then
11   if  $|CS| = 0$  then
12      $CS_0 = v$ ;
13      $|CS| = 1$ ;
14     send a lane-changing command to  $CS_0$  if  $CS_0$ 
        is not on the continuous lane;
15   else
16      $i = |CS| + 1$ ;
17      $CS_i = v$ ;
18      $|CS| = |CS| + 1$ ;
19   end
20 end
21 return  $CS$ ,  $DS$ ;
```

, and the lateral model for leader vehicle L in the k -th MVC is designed as

$$\begin{cases} \dot{q}_L^k(t) = \theta_L^k(t) \\ \dot{\theta}_L^k(t) = -d_{LL'}^k \left[\psi_3 (q_L^k(t) - q_L^{k-1} - r_{LL'q}^k) - \xi_3 (\theta_L^k(t) - \theta_L^{k-1}(t)) \right] \end{cases} \quad (7)$$

where $d_{LL'}^k$ and e_{Lm}^k are the adjacency weight between leader vehicle L and leader vehicle L' , follower vehicle m in the k -th MVC, respectively; $r_{LL'}^k = [r_{LL'p}^k, r_{LL'q}^k]^T$ is the desired gap between leader vehicle L in the k -th MVC and leader vehicle L' in the $k-1$ -th MVC, $r_{LL'p}^k, r_{LL'q}^k$ are the desired longitudinal gap and lateral gap; $\mathbf{x}_m^o = [p_m^o, q_m^o]^T$ is the position of vehicle m in the o -th MVC; $\boldsymbol{\psi} = [\psi_1, \psi_2, \psi_3]^T$ and $\boldsymbol{\xi} = [\xi_1, \xi_2, \xi_3]^T$ are the gain parameters. The meanings of other notations are similar to those defined in formula (3).

Remark 4: Aiming at preventing collisions in the process of forming a stable platoon, the leader vehicle can adaptively select reference vehicles according to its own state, i.e., acceleration or deceleration. If the leader vehicle L is in the acceleration state, then $o = k-1$ and m is the last follower vehicle of the o -th MVC; If the leader vehicle L is in the deceleration state, then $o = k$ and m is the first follower vehicle of the o -th MVC.

B. Platoon Control Algorithms in SD-M3ASP

This section elaborates on the algorithms mentioned in Workflow 1 separately.

1) *Topology Construction Algorithm:* When a vehicle v passes p_s , Algorithm 1 determines the role of v (i.e., leader or follower) and construct the topology of v (i.e., MVC).

Algorithm 2: Adaptive Topology Switching

Input: DS
Output: DS

```

1 if  $p_{\mathcal{H}}^{M^1} > p_d$  then
2   Obtain the maximum allowable capacity of the
        MVC  $\hat{n}$  at continuous lane via formula (8);
3   Obtain the average cluster capacity of MVCs  $\bar{n}$  via
        formula (9);
4   if  $\bar{n} \leq \hat{n}$  then
5     return Algorithm 3;
6   else
7     return Algorithm 4;
8   end
9 else
10  Send keep-driving commands to vehicles in  $DS$ ;
11  return  $DS$ ;
12 end
```

Specifically, if the longitudinal position of the leader vehicle of CS is greater than p_c , the existing MVC is packed into DS as a unique sub-decision set M^k for subsequent platoon control (lines 1-6 in Algorithm 1). Otherwise, an MVC with leader vehicle v is generated, and v moves to the continuous lane if CS is currently an empty set (lines 11-14 in Algorithm 1), or else vehicle v join in the existing MVC as the role of follower (lines 15-19 in Algorithm 1). Vehicles in CS are required to keep driving with the same velocity as the leader vehicle in CS (lines 7-9 in Algorithm 1).

Remark 5: Calculation errors may lead to multiple vehicles at position p_s simultaneously, thus the vehicle in the continuous lane has a higher priority to become the leader CS_0 .

Remark 6: In order to implement MVC in the proposed SD-M3ASP architecture, we design two logical forms of MVC, i.e., CS and DS . Specifically, CS is responsible for the initial generation of MVC in the range of (p_s, p_c) . Any vehicle will be assigned to a CS , which will be stored in DS as M^k . DS is mainly responsible for evaluating the real-time lane capacity and preprocessing M^k , and M^k is the smallest decision-making unit of the platoon control.

2) *Adaptive Topology Switching Algorithm:* After obtaining the topology of MVC according to Algorithm 1, Algorithm 2 first evaluate two metrics ($\hat{n}$, $\bar{n}$) of the road condition within the range of $(p_c, p_e]$ (lines 2-3 in Algorithm 2), which are defined in formulas (8) and (9), respectively.

$$\hat{n} = \left\lfloor \frac{p_{\mathcal{H}}^{M^1} - p_r^{M^{-1}}}{|DS|(F+Z)} \right\rfloor \quad (8)$$

$$\bar{n} = \frac{\sum_{m=1}^{|DS|} |M^m|}{|DS|} \quad (9)$$

Specifically, $\hat{n}$ is the maximum allowable capacity of continuous lane, which denotes the upper bound of vehicle number within the range from the head leader vehicle $\mathcal{H}$ in the first MVC M^1 to the rear follower vehicle r in the last MVC M^{-1} ; $\bar{n}$ is the average cluster capacity of MVCs, which denotes the average vehicle number per MVC in DS .

Algorithm 3: Sibling Topology Switching

Input: $DS, \hat{n}$
Output: DS

```

1 for  $k=1$  to  $|DS|$  do
2   foreach adjacent set  $M^j$  of  $M^k$  that satisfy
    $|M^k| > \hat{n}$ ,  $|M^j| < \hat{n}$  and  $|j - k| = 1$  do
3     if  $j - k < 0$  then
4       Assign priorities to vehicles of  $M^k$  in the
       terminated lane with positive order;
5     else
6       Assign priorities to vehicles of  $M^k$  in the
       continuous lane with the reverse order;
7     end
8     Transfer  $\min(|M^k| - \hat{n}, \hat{n} - |M^j|)$  vehicles
     from  $M^k$  to  $M^j$  according to priorities;
9   end
10 end
11 return  $DS$ 

```

Algorithm 4: Master-slave Topology Switching

Input: $DS, \hat{n}$
Output: DS

```

1 for  $k = 1$  to  $|DS|$  do
2   if  $|M^k| > \hat{n}$  then
3     Assign priorities to vehicles of  $M^k$  in the
     continuous lane with the reverse order;
4     Transfer  $|M^k| - \hat{n}$  vehicles from  $M^k$  to  $M^{k+1}$ 
     according to priorities;
5   end
6 end
7 return  $DS$ 

```

When the traffic flow is relatively sparse (i.e., $\bar{n} \leq \hat{n}$), it indicates that there are abundant spare spaces between and within MVCs. Therefore, evaluating each adjacent MVC pair assigns redundant vehicles in dense MVC to sparse MVC. Specifically, suppose the adjacent j -th MVC has a smaller capacity and is in front of the k -th MVC. In this case, the follower vehicle of the k -th MVC in the terminated lane which is relatively closer to the rear follower vehicle of the j -th MVC has a higher priority (lines 3-4 in Algorithm 3); otherwise, the follower vehicle of the k -th MVC in the continuous lane which is relatively farther away from the leader vehicle of the k -th MVC has a higher priority (lines 5-7). Thus, the k -th MVC has $|M^k| - \hat{n}$ redundant vehicles and the j -th MVC has $\hat{n} - |M^j|$ safe vacancies. Accordingly, $\min()$ operation is adopted, thus $\min(|M^k| - \hat{n}, \hat{n} - |M^j|)$ vehicles with higher priorities are taken out and transferred from M^k to M^j (line 8 in Algorithm 3).

When the traffic flow is relatively dense (i.e., $\bar{n} > \hat{n}$), it indicates that there is barely any space between and within MVCs. We demonstrate that it is necessary to provide spare spaces for the formation of vehicles ahead as early as possible, signifying that the congestion bottleneck is compulsorily back propagated to subsequent vehicles. Therefore, the dense MVC

Algorithm 5: Adaptive Platoon Control

Input: DS
Output: DS

```

1 for  $k = 1$  to  $|DS|$  do
2   Assess the status of the leader vehicle in  $M^k$  via
   metric (10);
3   Update the longitudinal states of the leader vehicle
   in  $M^k$  via formula (6);
4   if the distance between  $M_0^k$  and  $M_0^{k-1}$  are greater
   than  $|M^k|(F + Z)$ , and  $M_0^k$  and  $M_0^{k-1}$  have the
   same longitudinal velocity/acceleration then
5     Append vehicle  $M_0^{k+1}$  in  $M^k$ ;
6     Update the longitudinal states of follower
     vehicles in  $M^k$  via formula (4);
7   else
8     Update the longitudinal states of follower
     vehicles in  $M^k$  via formula (3);
9   end
10  if the longitudinal distances of vehicles meet
   collision avoidance conditions then
11    Update the lateral states of the leader vehicle in
     $M^k$  via formula (7);
12    Update the lateral states of follower vehicles in
     $M^k$  via formula (5);
13  end
14 end
15 return  $DS$ 

```

is promptly discovered (lines 1-2 in Algorithm 4) to assign redundant vehicles to the subsequent MVC. Specifically, if the k -th MVC has high capacity, then the follower vehicle of the k -th MVC assigns the priorities (line 3 in Algorithm 4) according to the same rule in line 6 of Algorithm 3. Thus, the k -th MVC assign $|M^k| - \hat{n}$ redundant vehicles to M^j (line 4 in Algorithm 4).

3) *Adaptive Platoon Control Algorithm:* We follow a simple rationale that longitudinal controller precedes lateral controller and leader precedes followers. As demonstrated in Algorithm 5, we update the longitudinal states of leaders, the longitudinal states of followers, the lateral states of leaders, and the lateral states of followers in lines 3, 6 and 8, 11, and 12, respectively.

Before updating the longitudinal states of leaders, the assessment of the leader in each MVC is necessary for preventing collisions. Thus, we introduce a metric as shown in formula (10) to judge whether the leader vehicle of M^k accelerates, decelerates, or keeps a constant speed.

$$\frac{p_r^{M^{k-1}} - p_{\mathcal{H}}^{M^k}}{F + Z} > 1 \wedge \frac{p_{\mathcal{H}}^{M^1} - p_{\mathcal{H}}^{M^k}}{\sum_m |DS| |M^m|(F + Z)} > 1 \quad (10)$$

Specifically, metric (10) combines actual and ideal spacing quantification. The first assertion is the left part of metric (10), which quantifies the actual spacing between the leader vehicle $\mathcal{H}$ of M^k and the rear follower vehicle r of M^{k-1} . If the first assertion is greater than 1, it signifies that the leader

vehicle $\mathcal{H}$ of M^k is allowed to accelerate without collision with the front vehicle $\mathcal{r}$ of M^{k-1} . The second assertion is the right part of metric (10), which quantifies the ideal spacing between the leader vehicle $\mathcal{H}$ of M^k and the leader vehicle $\mathcal{H}$ of M^1 . If the second assertion is greater than 1, it signifies that even if the cluster in front of M^k does not form a stable platoon, the leader vehicle $\mathcal{H}$ of M^k can still accelerate after reserving the ideal spacing. When metric (10) is true, we set $o = k-1$ and m as the rear follower vehicle of M^o in formula (6); otherwise, we set $o = k$ and m as the first follower vehicle of M^o in formula (6). Then, leader vehicles update their longitudinal states via the HL-AF model in formula (6) (lines 2-3 in Algorithm 5).

Subsequently, if the distance between leader vehicles M_0^k and M_0^{k-1} fluctuates little, follower vehicles update their longitudinal states via the DL-FF model in formula (4) (lines 4-6 in Algorithm 5); otherwise, they update their longitudinal states via the traditional Leader-Follower model in formula (3) (lines 7-9 in Algorithm 5). Then, if the desired longitudinal gaps between vehicles are met, the lateral states of leader and follower vehicles are updated via the HL-AF in formula (7) and the DL-FF model in formula (5), respectively.

Finally, the leader and followers of an MVC can harmoniously form a stable platoon without collision, and MVC-based platoons transit smoothly from the terminated lane to continuous lane. Therefore, the road capacity can be significantly augmented and the congestion time can be conceivably reduced.

VI. CONVERGENCE ANALYSIS

This section provides the theoretical analysis of the cooperative system based on the proposed DL-FF and HL-AF models.

A. Convergence Analysis of Intra-platoon Kinematic Model

Theorem 1: Suppose that the communication topology of the intra platoon is connected and the gain parameters γ_1, γ_2 are positive. Then, the intra-platoon consensus of each platoon can be reached based on the proposed intra-platoon model (4) and (5), i.e.:

- $\lim_{t \rightarrow \infty} \|\mathbf{x}_i^k(t) - \mathbf{x}_{\mathcal{H}}^k(t) - r_{i\mathcal{H}}^k\| = \lim_{t \rightarrow \infty} \|\mathbf{x}_i^k(t) - \mathbf{x}_{\mathcal{R}}^k(t) - r_{i\mathcal{R}}^k\| = 0$
- $\lim_{t \rightarrow \infty} \|\mathbf{v}_i^k(t) - \mathbf{v}_{\mathcal{H}}^k(t)\| = \lim_{t \rightarrow \infty} \|\mathbf{v}_i^k(t) - \mathbf{v}_{\mathcal{R}}^k(t)\| = 0$

Proof: If $c_{i\mathcal{R}}^k = 0$, define the position error and velocity error as

$$\begin{aligned} \tilde{p}_i^k(t) &= p_i^k(t) - p_{\mathcal{H}}^k(t) - r_{i\mathcal{H}}^k \\ \tilde{\omega}_i^k(t) &= \omega_i^k(t) - \omega_{\mathcal{H}}^k(t) \end{aligned} \quad (11)$$

If $c_{i\mathcal{R}}^k \neq 0$, then let $b_i^k = c_i^k$. Define the position error and velocity error as

$$\begin{aligned} \tilde{p}_i^k(t) &= p_i^k(t) - p_{\mathcal{H}}^k(t) - r_{i\mathcal{H}}^k + p_i^k(t) - p_{\mathcal{R}}^k(t) - r_{i\mathcal{R}}^k \\ \tilde{\omega}_i^k(t) &= \omega_i^k(t) - \omega_{\mathcal{H}}^k(t) + \omega_i^k(t) - \omega_{\mathcal{R}}^k(t) \end{aligned} \quad (12)$$

According to formulas (11) and (12), formula (4) can be rewritten as

$$\begin{cases} \dot{\tilde{p}}_i^k(t) = \tilde{\omega}_i^k(t) \\ \dot{\tilde{\omega}}_i^k(t) = - \sum_{j=1}^N a_{ij}^k [\phi_p(\tilde{p}_i^k(t) - \tilde{p}_j^k(t)) + \zeta_{\omega}(\tilde{\omega}_i^k(t) - \tilde{\omega}_j^k(t))] \\ \quad - b_{ij}^k [\phi_p \tilde{p}_i^k(t) + \zeta_{\omega} \tilde{\omega}_i^k(t)] \end{cases} \quad (13)$$

Then, we define:

$$\begin{aligned} p^k(t) &= [\tilde{p}_1^k(t), \dots, \tilde{p}_n^k(t)]^r \\ \omega^k(t) &= [\tilde{\omega}_1^k(t), \dots, \tilde{\omega}_n^k(t)]^r \\ \sigma^k(t) &= [p^k(t), \omega^k(t)]^r \end{aligned} \quad (14)$$

It follows from (13) that:

$$\begin{aligned} \dot{\sigma}^k(t) &= \begin{pmatrix} 0_{n \times n} & I_{n \times n} \\ -\phi_p(L+K) & -\zeta_{\omega}(L+K) \end{pmatrix} \sigma^k(t) \\ &= \begin{pmatrix} 0_{n \times n} & I_{n \times n} \\ -\phi_p H & -\zeta_{\omega} H \end{pmatrix} \sigma^k(t) = \Sigma \sigma^k(t) \end{aligned} \quad (15)$$

where $H = L + K$, $\Sigma = \begin{pmatrix} 0_{n \times n} & I_{n \times n} \\ -\phi_p H & -\zeta_{\omega} H \end{pmatrix}$ and the matrices L and K are defined in Section III.

Thereby, the intra-platoon convergence analysis is converted into the stability analysis of formula (15), i.e., we only need to prove that all the eigenvalues of Π are in the left of the complex plane.

Let the eigenvalue of H be: $h_1, \dots, h_n$. Since Hermite matrix H is symmetric, there exists a unitary matrix U to satisfy

$$U^{-1} H U = \Lambda = \text{diag}(h_1, \dots, h_n) \quad (16)$$

To facilitate the convergence analysis, the following characteristics of matrix H is introduced.

Lemma 1: [43] The eigenvalues of matrix H are non-negative real numbers. Besides, the eigenvalues of matrix H are all positive real numbers if and only if the graph is connected.

Since the communication graph of follower vehicles is connected, according to Lemma 1, we know that all eigenvalues of the matrix H are nonnegative real numbers, i.e., $h_i > 0$.

According to formulas (15) and (16), we have:

$$\begin{aligned} \det(\rho I_{2n \times 2n} - \Sigma) &= \det[\rho^2 I_{n \times n} + \rho \zeta_{\omega}(U^{-1} H U) + \phi_p(U^{-1} H U)] \\ &= \det[\rho^2 I_{n \times n} + \rho \zeta_{\omega} \Lambda + \phi_p \Lambda] \\ &= \prod_{i=0}^n (\rho^2 + \rho \zeta_{\omega} h_i + \phi_p h_i) \end{aligned} \quad (17)$$

where $\det(\cdot)$ represents the determinant of the matrix. Besides, each h_i corresponds to two eigenvalues ρ_i of Σ as

$$\rho_{i\pm} = \frac{-\zeta_{\omega} \pm \sqrt{\zeta_{\omega}^2 h_i^2 - 4\phi_p h_i}}{2} \quad (18)$$

where the gaining parameters ϕ_p, ζ_{ω} are greater than 0.

According to the Routh-Hurwitz stability criterion [44], we have the real part of characteristic polynomial $f(\rho_i) = \rho_i^2 + \zeta_{\omega} h_i \rho_i + \phi_p h_i$'s root is in the left half plane, i.e., $\text{Re}(\rho_{i\pm}) < 0$. Therefore, formula (15) is asymptotically stable based on Lyapunov stability theorem. Then, we have

$\lim_{t \rightarrow \infty} \|\sigma^k(t) - \sigma_e^k(t)\| = 0$, where $\sigma_e^k(t) = [0, 0]^T$. Hence, we know that $\lim_{t \rightarrow \infty} \|\sigma^k(t)\| = 0$. According to formula (14), $\lim_{t \rightarrow \infty} \|\tilde{p}^k(t)\| = \lim_{t \rightarrow \infty} \|\tilde{\omega}^k(t)\| = 0$ can be obtained.

Since the longitudinal model (4) and lateral model (5) only differ in $c_{i\mathcal{R}}^k$, which is summarized in formula (11) and (12), the lateral convergence is obviously the same as the longitudinal convergence. Accordingly, we have $\lim_{t \rightarrow \infty} \|\mathbf{x}_i^k(t) - \mathbf{x}_L^k(t) - r_{i\mathcal{R}}^k\| = \lim_{t \rightarrow \infty} \|\mathbf{x}_i^k(t) - \mathbf{x}_L^k(t) - r_{i\mathcal{R}}^k\| = \lim_{t \rightarrow \infty} \|\tilde{\mathbf{x}}^k(t)\| = 0$, $\lim_{t \rightarrow \infty} \|\mathbf{v}_i^k(t) - \mathbf{v}_L^k(t)\| = \lim_{t \rightarrow \infty} \|\tilde{\mathbf{v}}^k(t)\| = 0$.

This completes the proof of Theorem 1. ■

B. Convergence Analysis of Inter-platoon Kinematic Model

Theorem 2: Suppose that the communication topology of the inter platoon is connected. Then, the inter-platoon consensus among platoons can be reached based on the proposed inter-platoon model (6) and (7), i.e.:

- $\lim_{t \rightarrow \infty} \|\mathbf{x}_L^k(t) - \mathbf{x}_L^1(t) - r_{L1}^k\| = \lim_{t \rightarrow \infty} \|\mathbf{x}_L^k(t) - \mathbf{x}_m^d(t) - r_{Lm}^k\| = 0$;
- $\lim_{t \rightarrow \infty} \|\mathbf{v}_L^k(t) - \mathbf{v}_L^1(t)\| = \lim_{t \rightarrow \infty} \|\mathbf{v}_L^k(t) - \mathbf{v}_m^d(t)\| = 0$

Proof: The design idea of our **platoon** is to refer to the leader of the $k-1$ -th MVC in the multi-agent mobile-edge computing networks, which is easy to deploy but non-trivial to analyze the convergence. Thereby, for the convenience of proof, our reference vehicles are all leader vehicles in the first MVC. Thus, we combine longitudinal and lateral models (6) and (7) as

$$\begin{cases} \dot{\mathbf{x}}_L^k(t) = \mathbf{v}_L^k(t) \\ \dot{\mathbf{v}}_L^k(t) = -\psi(\mathbf{x}_L^k(t) - \mathbf{x}_L^1(t) - r_{L1}^k) - \xi(\mathbf{v}_L^k(t) - \mathbf{v}_L^1(t)) \\ \quad - \psi(\mathbf{x}_L^k(t) - \mathbf{x}_m^d(t) - r_{Lm}^k) - \xi(\mathbf{v}_L^k(t) - \mathbf{v}_m^d(t)) \end{cases} \quad (19)$$

where $\mathbf{x}_L^1$ is the position of the leader vehicle in the first MVC; r_{L1} is the desired gap between the leader of the current MVC and the leader of the first MVC.

Then, the inter-platoon convergence is analyzed by solving the differential equation shown in formula (19).

Firstly, we need to prove that a stable platoon can be formed between the leader vehicles of each MVC, i.e., $\mathbf{v}_L^k(t) = -\psi(\mathbf{x}_L^k(t) - \mathbf{x}_L^1(t) - r_{L1}^k) - \xi(\mathbf{v}_L^k(t) - \mathbf{v}_L^1(t))$ tend toward stability. This formula can be regarded as the following inhomogeneous differential equation:

$$\ddot{\mathbf{x}}_L^k(t) + \xi\dot{\mathbf{x}}_L^k(t) + \psi\mathbf{x}_L^k(t) = \dot{\mathbf{x}}_L^1(t) + \xi\dot{\mathbf{x}}_L^1(t) + \psi(\mathbf{x}_L^1(t) - r_{L1}^k) \quad (20)$$

Let the general solution of the above formula (20) be as

$$\mathbf{x}_L^k(t) = (\mathbf{x}_L^k)^*(t) + c_L^k X_L^k(t) \quad (21)$$

where $(\mathbf{x}_L^k)^*(t)$ is the particular solution of formula (20). It can be derived according to formula (20) as

$$(\mathbf{x}_L^k)^*(t) = \mathbf{x}_L^1(t) + r_{L1}^k \quad (22)$$

According to formula (22), the general solution (21) can be rewritten as

TABLE III: The initial states and desired gaps of intra-platoon simulations

Idx.	Position (m)	Velocity (m/s)	Desired gap L1 $\leftrightarrow$ F1-F4 (m)	Desired gap L2 $\leftrightarrow$ F1-F4 (m)
L1	(60, 1) ^T	(10, 0) ^T	0	-
F1	(40, 1) ^T	(10, 0) ^T	(10, 0) ^T	(-40, 0) ^T
F2	(40, 2) ^T	(10, 0) ^T	(20, 0) ^T	(-30, 0) ^T
F3	(30, 1) ^T	(10, 0) ^T	(30, 0) ^T	(-20, 0) ^T
F4	(30, 2) ^T	(10, 0) ^T	(40, 0) ^T	(-10, 0) ^T
L2	(10, 1) ^T	(10, 0) ^T	-	0

$$\mathbf{x}_L^k(t) = \mathbf{x}_L^1(t) + r_{L1}^k + c_L^k X_L^k(t) \quad (23)$$

where c_L^k is the constant coefficient and $X_L^k(t)$ is the general solution.

For different values of ψ and ξ , $X_L^k(t)$ can be divided into the following three cases:

- If $\xi^2 - 4\psi > 0$, then:

$$\mathbf{x}_L^k(t) = (c_L^k)_{11}e^{\lambda_1 t} + (c_L^k)_{12}e^{\lambda_2 t} \quad (24)$$

where $\lambda_{1,2} = \frac{-\xi \pm \sqrt{\xi^2 - 4\psi}}{2} < 0$; $(c_L^k)_{11}$ and $(c_L^k)_{12}$ are constant coefficients.

- If $\xi^2 - 4\psi = 0$, then:

$$\mathbf{x}_L^k(t) = ((c_L^k)_{21} + (c_L^k)_{22}t)e^{\lambda_1 t} \quad (25)$$

where $\lambda_{1,2} = \frac{-\xi}{2} < 0$; $(c_L^k)_{21}$ and $(c_L^k)_{22}$ are constant coefficients.

- If $\xi^2 - 4\psi < 0$, then:

$$\mathbf{x}_L^k(t) = e^{\alpha t}((c_L^k)_{31} \cos(\beta t) + (c_L^k)_{32} \sin(\beta t)) \quad (26)$$

where $\alpha = \frac{-\xi}{2} < 0$; $\beta = \frac{\sqrt{4\psi - \xi^2}}{2}$; $(c_L^k)_{31}$ and $(c_L^k)_{32}$ are constant coefficients.

Based on formulas (24), (25) and (26), we have:

$$\lim_{t \rightarrow \infty} \|\mathbf{x}_L^k(t)\| = 0 \quad (27)$$

Furthermore, we can obtain from formula (23) that:

$$c_L^k X_L^k(t) = \mathbf{x}_L^k(t) - \mathbf{x}_L^1(t) - r_{L1}^k \quad (28)$$

Based on formulas (27) and (28), we have $\lim_{t \rightarrow \infty} \|\mathbf{x}_L^k(t) - \mathbf{x}_L^1(t) - r_{L1}^k\| = \lim_{t \rightarrow \infty} \|c_L^k X_L^k(t)\| = 0$ and $\lim_{t \rightarrow \infty} \|\mathbf{v}_L^k(t) - \mathbf{v}_L^1(t)\| = \lim_{t \rightarrow \infty} \|\dot{c}_L^k X_L^k(t)\| = 0$. Similarly, we can derive that $\lim_{t \rightarrow \infty} \|\mathbf{x}_L^k(t) - \mathbf{x}_m^d(t) - r_{Lm}^k\| = \lim_{t \rightarrow \infty} \|c_L^k X_L^k(t)\| = 0$ and $\lim_{t \rightarrow \infty} \|\mathbf{v}_L^k(t) - \mathbf{v}_m^d(t)\| = \lim_{t \rightarrow \infty} \|\dot{c}_L^k X_L^k(t)\| = 0$.

This completes the proof of Theorem 2. ■

VII. NUMERICAL SIMULATIONS

In this section, the numerical simulation results are given to verify the intra-platoon and inter-platoon convergence of the proposed models in Section VII-A and VII-B, respectively. Additionally, the comprehensive comparative simulations between SD-M3ASP and some traditional lane-changing & car-following models are given in Section VII-C to demonstrate

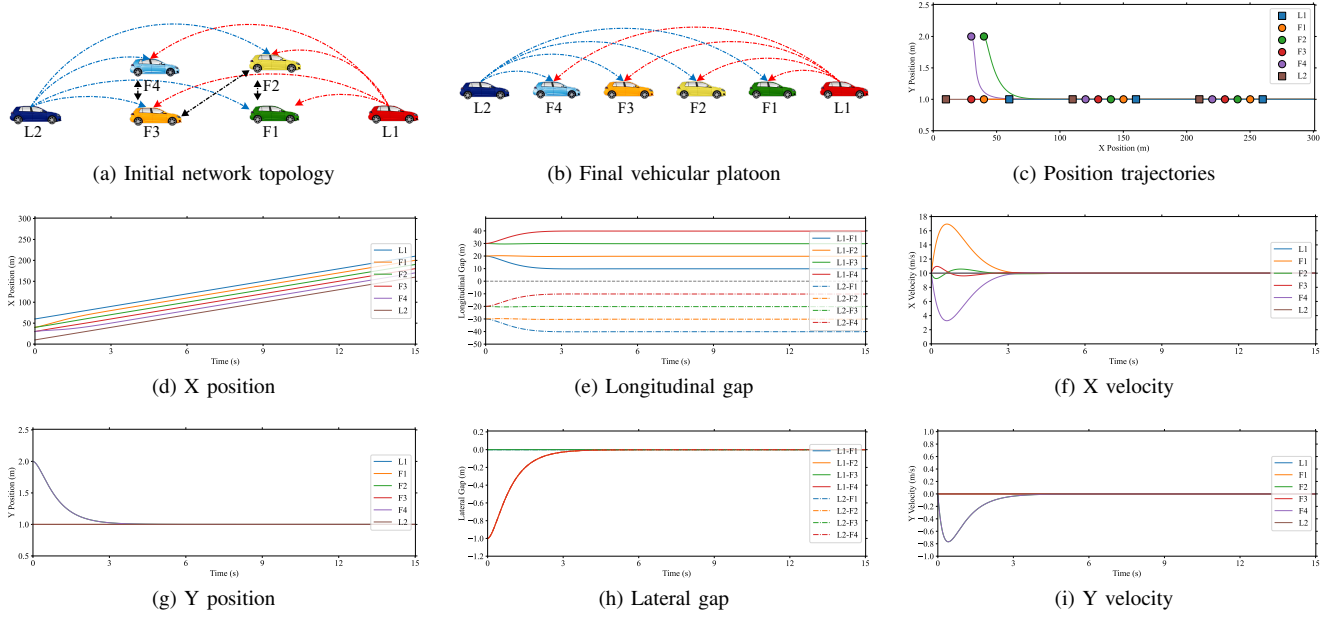


Fig. 5: The intra-platoon vehicular trajectories

the superiority of our method. The proposed methods are implemented by both traditional traffic simulation platform SUMO-1.12.0 [45] and our developed simulation platform VSVP (Visual Simulation of Vehicular Platoon). Due to space limit, we refer readers to our developed system¹ for more details of the simulations.

A. Validation of the intra-platoon model

In this section, the effectiveness of the proposed intra-platoon model in formulas (4) and (5) is validated. The adjacency weights are set to 1, i.e., $b_{i\mathcal{H}}^k = 1$ and $c_{i\mathcal{R}}^k = 1$. The gain parameters are set to 1 for the longitudinal controller and 5 for the lateral controller to make the lateral offset more sensitive, i.e., $\phi = [1, 1, 1, 5]^T$ and $\zeta = [1, 1, 1, 5]^T$. The initial topology network and final vehicular platoon are shown in Figs. 5a and 5b, respectively. Besides, the initial states and desired distances are given in TABLE III.

Fig. 5c depicts the two-dimensional position trajectories of all vehicles. Leaders L1 and L2 keep the moving direction and followers F1-F4 can form a stable platoon with them in a finite time. Specifically, leader and follower vehicles form an equidistant stable platoon as shown in Fig. 5d, and the longitudinal gaps between them converge to the desired gaps as shown in 5e, and the longitudinal velocities of them converge to the desired velocity of leader L1 as shown in Fig. 5f. Meanwhile, leader and follower vehicles move to the same Y coordinate as shown in Fig. 5g, and the lateral gaps between them converge to zero as shown in Fig. 5h, and the lateral velocities converge to zero as shown in Fig. 5i.

Consequently, vehicles can asymptotically reach a intra-platoon consensus according to the proposed DL-FF model of formulas (4) and (5).

B. Validation of the inter-platoon model

In this section, the effectiveness of the proposed intra-platoon model in formulas (6) and (7) is validated. The adjacency weights are set to 1, i.e., $d_{LL'}^k = 1$ and $e_{Lm}^k = 1$. The gain parameters are set to 1 for the longitudinal controller and 5 for the lateral controller to make the lateral offset more sensitive, i.e., $\psi = [1, 1, 5]^T$ and $\xi = [1, 1, 5]^T$. The initial topology network and final vehicular platoon are shown in Figs. 6a and 6b, respectively. **Notably, there are initially two MVCs, i.e., an MVC consisting of L1, F1, and F2, and the other MVC consisting of L2, F3, and F4.**

Besides, the initial states and desired distances are given in TABLE IV.

Fig. 6c depicts the two-dimensional position trajectories of all vehicles. **The MVC of leader L2 can keep pace with the MVC of leader L1, thus forming two stable platoons in a finite time.** Specifically, vehicles of each MVC form an equidistant stable platoon as shown in Fig. 6d, and the longitudinal gaps between follower vehicles of each MVC and their corresponding leaders converge to the desired gaps as shown in 6e, and the longitudinal velocities of them converge to the desired velocity of leader L1 as shown in Fig. 6f. Meanwhile, leader and follower vehicles move to the same Y coordinate as shown in Fig. 6g, and the lateral gaps between them converge to zero as shown in Fig. 6h, and the lateral velocities converge to zero as shown in Fig. 6i.

Consequently, MVC-based platoons can asymptotically reach an inter-platoon consensus according to the proposed HL-AF model of formulas (6) and (7).

C. Comprehensive comparisons between the SD-M3ASP architecture and the baseline methods

To demonstrate the superiority of the proposed algorithms and models in the SD-M3ASP architecture, we use the fol-

¹<http://qgailab.com/cav-sim>

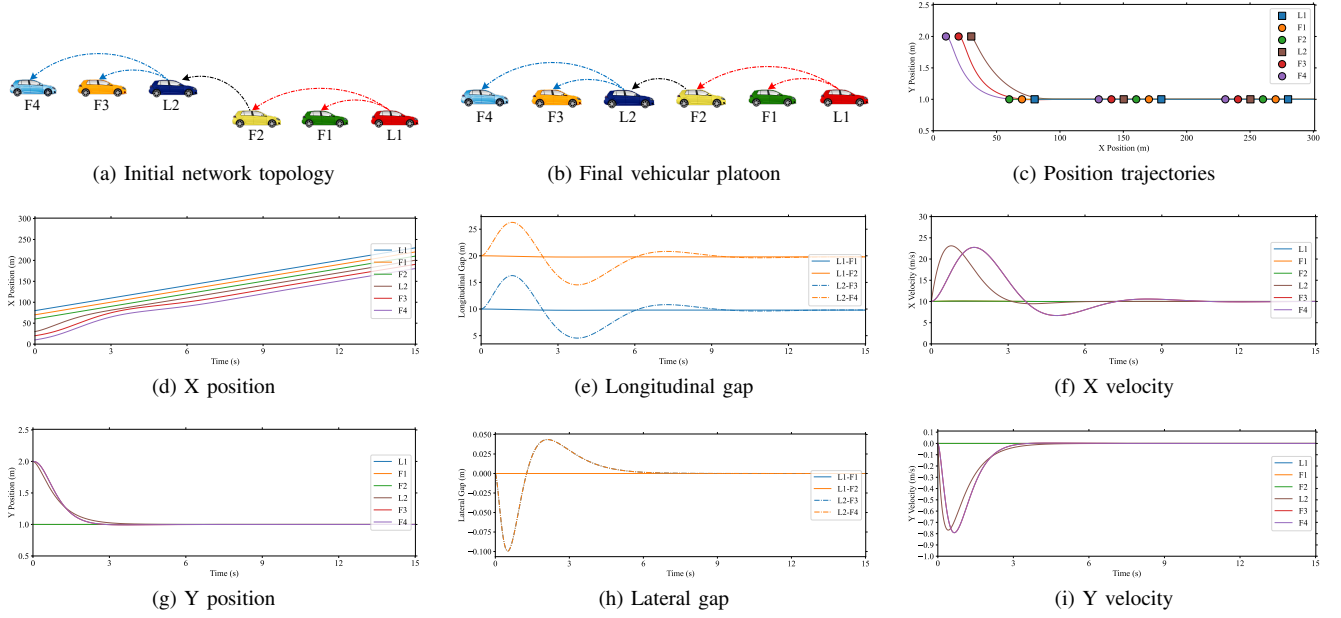


Fig. 6: The inter-platoon vehicular trajectories

TABLE IV: The initial states and desired gaps of inter-platoon simulations

Idx.	Position (m)	Velocity (m/s)	Desired gap			
			L1 ↔ F1,F2 (m)	L2 ↔ F3,F4 (m)	L2 ↔ L1 (m)	L2 ↔ F2 (m)
L1	$(80, 1)^T$	$(10, 0)^T$	-	-	$(-30, 0)^T$	-
F1	$(70, 1)^T$	$(10, 0)^T$	$(10, 0)^T$	-	-	-
F2	$(60, 1)^T$	$(10, 0)^T$	$(20, 0)^T$	-	-	$(-10, 0)^T$
L2	$(30, 2)^T$	$(10, 0)^T$	-	-	-	-
F3	$(20, 2)^T$	$(10, 0)^T$	-	$(10, 0)^T$	-	-
F4	$(10, 2)^T$	$(10, 0)^T$	-	$(20, 0)^T$	-	-

lowing four metrics:

- $\mathcal{T}$ [46] measures the average travel time (s) of N vehicles. $\mathcal{T}_i$ is the travel time from p_s to p_e of vehicle i .

$$\mathcal{T} = \frac{1}{N} \sum_i \mathcal{T}_i \quad (29)$$

- $\mathcal{C}$ [47] measures the total traffic capacity of K MVCs. $\mathcal{C}_k$ (veh/s) is the number of vehicles of k -th MVC in unit time. $\bar{\omega}$ is the stable longitudinal velocity of k -th MVC. F' is the inter-platoon spacing between the head leader vehicle $\mathcal{H}$ of k -th MVC and the rear follower vehicle $\mathcal{r}$ of $k-1$ -th MVC. The bigger value of $\mathcal{C}$ indicates the denser the traffic flow and the fewer the traffic congestion.

$$\mathcal{C} = \sum_k \mathcal{C}_k \quad (30)$$

$$\mathcal{C}_k = \bar{\omega} \frac{|D^k|}{|D^k|Z + (|D^k| - 1)F + F'} \quad (31)$$

- $\mathcal{P}$ measures the average percentage (%) of total early lane change. $\mathcal{P}_i$ is the lane-changing position (m) of

vehicle i from the terminated lane to the continuous lane. The smaller value of $\mathcal{P}$ indicates the earlier change to the continuous lane and the faster the stability.

$$\mathcal{P} = \frac{1}{N} \sum_i \left(\frac{\mathcal{P}_i - p_s}{p_e - p_s} \right) \quad (32)$$

- $\mathcal{E}$ [49] measures the total pollutant emission (g) of all vehicles in their running time of $[p_s, p_e]$. $\mathcal{E}_i(t)$ is the emission (g/s) of vehicle i at time t . $\mathcal{E}_0$ is a lower bound of emission, and f_1 to f_6 are regression coefficients of the specific pollutant type and vehicle type. We focus on the CO₂ pollutant in the simulations, thus $\mathcal{E}_0 = 0$, $f_1 = 3.24e-01$, $f_2 = 8.59e-02$, $f_3 = 4.96e-3$, $f_4 = -5.86e-2$, $f_5 = 4.48e-1$, and $f_6 = 2.30e-1$.

$$\mathcal{E} = \sum_i \sum_t \mathcal{E}_i(t) \quad (33)$$

$$\mathcal{E}_i(t) = \max(\mathcal{E}_0, f_1 + f_2 v_i(t) + f_3 v_i(t)^2 + f_4 a_i(t) + f_5 a_i(t)^2 + f_6 v_i(t) a_i(t)) \quad (34)$$

Additionally, two baseline methods [50] (*IDM+SL* and *Krauss+SL*) are used for comprehensive comparisons. The naming format of these methods is: 'car-following model'+ 'lane-changing model'. The settings of common parameters used in this section are given in TABLE V. We demonstrate the comparison results in Fig. 7, where the x -axis and y -axis represents the emit probability of a vehicle in the continuous lane and the terminated lane, respectively, and the z -axis represent the value of the specific evaluation metric.

- Figs. 7a, 7e, and 7i depict the results of $\mathcal{T}$: With the increase of p_1 and p_2 , the travel time of traditional methods is growing higher, showing the shape of a "Sand Dune", while the travel time of SD-M3ASP is stable,

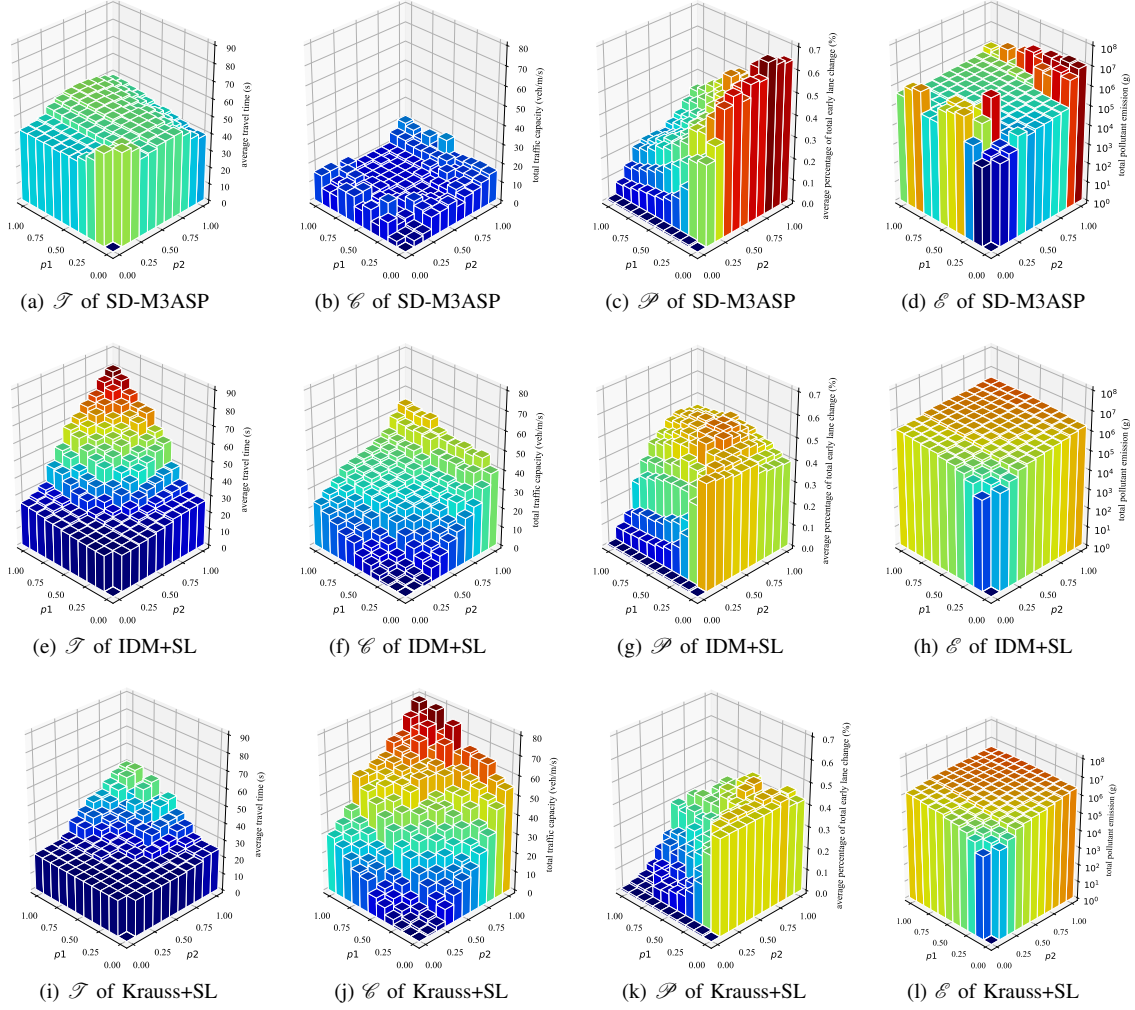


Fig. 7: The comparison results between the SD-M3ASP architecture and the baseline methods

TABLE V: Parameter Settings of Comparative Simulations

Sym.	Value	Unit	Meaning
p_s	0	m	the start position
p_c	100	m	the end of Topology Construction Stage
p_d	300	m	the end of Topology Switching Stage
p_e	500	m	the position of the terminated lane
Z	4.2	m	the secure distance
W	1.8	m	the width of vehicles
F	1.2	m	the length of vehicles
v_{init}	$[8.333, 0]^T$	m/s	the initial speed of vehicles
v_{max}	$[27.778, 8.333]^T$	m/s	the max speed of vehicles
$\dot{v}_{\text{max}}$	$[6, 4]^T$	m/s^2	the max acceleration of vehicles
p_1	0 to 100	%	the emit probability of an vehicle in the continuous lane (Lane 1)
p_2	0 to 100	%	the emit probability of an vehicle in the terminated lane (Lane 2)
$begin$	0	s	the departure time of the first vehicle
end	60	s	the end of departure interval
ϕ, ζ	$[5, 5, 5, 20]^T$	-	the gain parameters of DL-FF model
ψ, ξ	$[5, 5, 20]^T$	-	the gain parameters of HL-AF model

that is, it does not fluctuate with the increase of the emit probability. The reason is that our algorithm adjusts the

number of vehicles in the MVC in time according to the real-time traffic flow in the topology switching stage, and the kinematic model of the leader vehicle not only refers to the vehicle state of its own MVC but also refers to the leader vehicle state of the adjacent MVC. This makes the SD-M3ASP insensitive to emit probability. Therefore, SD-M3ASP has a strong generalization performance than the baseline methods.

- Figs. 7b, 7f, and 7j depict the results of $\mathcal{C}$: Compared with SD-M3ASP, the traditional methods have higher capacity, but the advantage of our method is that the capacity fluctuates less with the increase of p_1 and p_2 . This is because the leader vehicle of the first MVC in our method runs at a constant velocity, which can ensure that all MVCs form stable and safe platoons.
- Figs. 7c, 7g, and 7k depict the results of $\mathcal{P}$: The advantage of SD-M3ASP's early lane change rate can be gradually reflected when the density of the two lanes is increasing. Only when there are many vehicles in the terminated lane, the position of early lane change is relatively late. This is because our method needs to make the desired spacing in the longitudinal direction before

executing the lateral kinematic model. It is worth noting that the settings of p_c and p_d have a great impact on the early lane change probability of our method. Therefore, it should be set reasonably in combination with the road length and the speed limit in practical application.

- Figs. 7d, 7h, and 7l depict the results of $\mathcal{E}$: With the increase of p_1 and p_2 , the pollutant emission of SD-M3ASP is approximately an order of magnitude lower than that of the traditional methods. However, when p_2 is generally large, that is, there are more vehicles on the discontinuous lane, the pollutant emission of our method is higher than that of the traditional methods. This is because in the platoon control process of vehicles on the terminated lane, our algorithm needs to switch MVCs according to the real-time traffic flow, thus the acceleration and deceleration are more frequent.

Noticeably, **under the 2-to-1 lane reduction scenarios with heterogeneous traffic condition**, the proposed SD-M3ASP outperforms the baseline methods with more stable travel time, sufficient traffic capacity, faster lane change, and lower pollutant emission.

VIII. CONCLUSION

In this paper, in order to solve the problem of self-organized congestion with dynamic heterogeneity at the lane reduction bottleneck, we propose a software defined architecture for multi-vehicle adaptive platoon in the multi-agent mobile-edge computing networks, which is called SD-M3ASP. **We take the advantage of the beforehand platoon at the continuous lane to ensure efficient performance and high throughput based on SD-M3ASP.** The traffic flow capacity is considered to design the adaptive platoon control algorithms, in which the consensus-based intra- and inter-platoon kinematic models are provided. The theoretical analyses are carried out by using the Routh stability criterion. Extensive numerical experiments finally prove the effectiveness and superiority of the proposed SD-M3ASP.

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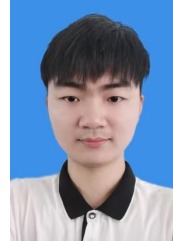
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